Dhruv Dave

Toronto, ON

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Education

Queen's University **Sep 2024 – Present**
Bachelor of Computer Science (Honours), Mathematics Minor *Kingston, ON*

- **Relevant Coursework**: Data Structures, Computer Architecture, System-Level Programming (C), Software Specifications, Linear Algebra; *in progress*: Operating Systems, Software Quality Assurance

Skills

Languages: Python, TypeScript, JavaScript, Java, C, SQL
Backend & APIs: FastAPI, Flask, Node.js, Express, Hono, SQLAlchemy, REST, JWT auth, Claude API
Data: PostgreSQL, Redis, SQLite, pandas, NumPy, ETL pipelines, Power BI
Frontend: React, Tailwind CSS, Chrome MV3
Delivery & Testing: Docker Compose, DigitalOcean, Linux, nginx, GitHub Actions (CI/CD), Vercel, Git, Bash, pytest, Vitest

Experience

Cerebro AI Technical Specialist | *FGF Brands* **May 2026 – Aug 2026**
Data & Platform Engineering *Toronto, ON*

- Traced silent Overall Equipment Effectiveness (OEE) reporting corruption at 22 bakery facilities to misconfigured workflow state transitions and ETL gaps in 2,000+ records with SQL diagnostics on FGF's Cerebro analytics platform
- Developed Node.js and Python tools documenting Cerebro's OEE telemetry models, changeover logic, and troubleshooting paths
- Shipped Power BI dashboards on Cerebro production data for plant teams tracking OEE, downtime, and changeover by site
- **Won 1st place at FGF's 4th annual FGFLife Hackathon** with BakeryInsight (React, Node.js, Claude API), which scores food trends from 9 sources and rates each on 6 feasibility factors against real FGF production-line and ingredient data

Research Assistant | *COBWEB, University of Toronto* **Jan 2024 – Aug 2024**
Computational Simulation & Data Analysis *Toronto, ON*

- Configured multi-condition agent-based experiments on the COBWEB platform (Dr. Brad Bass) that compared control and experimental populations in models of ecological adaptive systems
- Automated cleaning, normalization, and plotting of multi-run experiment output with pandas and NumPy pipelines, which the group used to compare population dynamics and threshold crossings

Software Engineer Intern | *Digitera* **Sep 2022 – Feb 2023**
Web Application Development *Toronto, ON*

- Delivered web applications for small-business clients in JavaScript, HTML, and CSS, and cut page load time ∼40% with asset minification, image compression, and deferred script loading

Projects

Algorithmic Trading Bot | *Python (asyncio), Interactive Brokers API, Docker Compose, DigitalOcean, Telegram*

- Built a trend-following system for US equities: EMA(12/26) crossovers behind a 200-SMA filter, ATR trailing stops, and a risk manager that sizes by equity at risk, halts on daily-loss or drawdown limits, and persists halt state across restarts
- Wrote a backtester that runs the live strategy code over 5 years of daily bars and reports return, drawdown, Sharpe, and win rate
- Fixed CAD/USD position-sizing errors with async FX-rate caching from IBKR forex data, and restricted signal evaluation to 3:50–4:00pm ET so the bot never trades on an unfinished daily bar
- Ran IB Gateway headless (IBC, Xvfb) beside the bot in Docker Compose on DigitalOcean for 24/7 paper trading, with Telegram commands to pause, check P&L, and close positions

Markr (Stock Screener) | *React 19, FastAPI, PostgreSQL, Redis, Docker* | github | live

- Put 6 market-data providers behind one FastAPI interface and rate-limited each with an atomic Redis Lua token bucket
- Range-partitioned price history by month in PostgreSQL and moved candle writes off the request path into a background task, with Redis TTLs set per resolution (60s intraday, 3600s daily)
- Covered the fallback chain, rate limiter, and indicator math with Vitest, with CI deploys to Vercel through GitHub Actions

ctx (Browser Session Tracker) | *TypeScript, Node.js, Hono, SQLite, Chrome MV3* | *Chrome Web Store* | github

- Published a local-first session tracker to the Chrome Web Store as a TypeScript monorepo (MV3 extension, Hono/Node.js backend, CLI) that stores sessions in `chrome.storage.local` with no server
- Added a content-script privacy guard that checks URL patterns, a sensitive-domain list, and live DOM state (password inputs, credit-card autocomplete) before capture and logs blocked pages without URL, title, or content

SecureVault: Password Security Analyzer & Manager | *React 18, Flask, Python, SQLite, Docker* | github | live

- Encrypted vault entries with AES-256-GCM under per-entry keys derived by PBKDF2 (600,000 iterations), and secured accounts with bcrypt (cost 12), JWT access/refresh rotation, TOTP 2FA, and login rate limiting
- Computed password strength 0–100 from entropy plus keyboard-walk, date-sequence, and common-pattern detection, with HaveIBeenPwned breach checks over SHA-1 prefix k-anonymity
- Containerized Flask and React behind nginx and covered auth, encryption, and breach checks with pytest